P34 - Visual Question Answering App

Generative AI - LLM Mini-Project Report

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Class / Batch	AIA-05 Batch B
Problem / Track	P34 - Visual Question Answering App (Track C (P25-P36: GPT, BERT and multimodal models))
Live app URL	https://huggingface.co/spaces/samjocodes/p34-vqa-app
GitHub repo	https://github.com/samjocodes/p34-vqa-app
Evaluation run	2026-11-07 14:50 | VQA: Salesforce/blip-vqa-base | LLM: Qwen/Qwen2.5-0.5B-Instruct (local) | device: cpu

Goal. Answer questions about an uploaded image. **Headline result:** the app answered **14/20 = 70.0%** of the 20 image-question pairs correctly (baseline), and **70.0%** when automatic brightness enhancement is applied to the low-light images.

1. System overview

The user uploads an image and asks a question, either typed or built from a question template (U1). **BLIP-VQA** (Salesforce/blip-vqa-base) produces a short answer with a confidence value (U3). A small LLM (Qwen/Qwen2.5-0.5B-Instruct, or the Gemini API if a key is configured) rewrites the terse answer as a complete sentence (U2). Guardrails refuse sensitive-attribute questions, flag low confidence and check that the expanded sentence still contains the original answer. A built-in test-image generator (U4) and evaluation tab (U5) produce the measurements in this report.

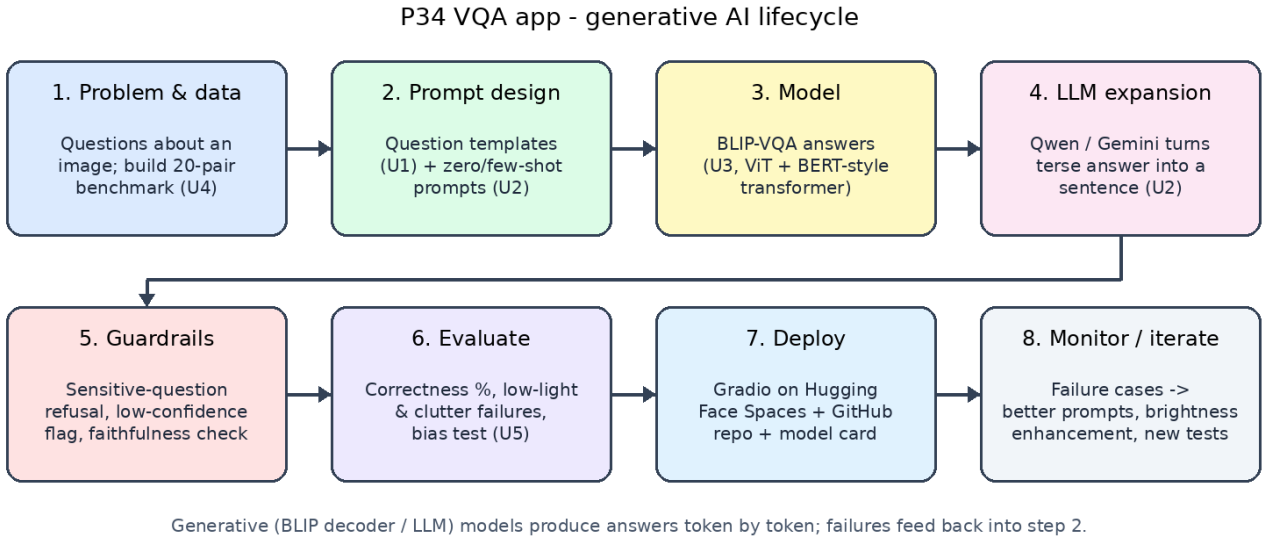


Figure 1. Project lifecycle: problem and data, prompt design, model, LLM expansion, guardrails, evaluation, deployment, iteration.

2. Unit-by-unit implementation (U1-U5)

Unit	Task in problem statement	What was built
U1	Create question templates	8 templates in 6 categories (colour, counting, existence, identity, location, scene/lighting) with fill-in-the-blank slots; question-type detector.
U2	LLM expands terse answers into full sentences	Expander with Gemini / Qwen/Qwen2.5-0.5B-Instruct / rule-based fallback; zero-shot vs few-shot prompts compared (Section 3).
U3	Answer with BLIP-VQA or a vision LLM	Salesforce/blip-vqa-base via Hugging Face Transformers (PyTorch); returns answer, mean token probability and top-5 first-token probabilities.
U4	Generate test images for hard cases	Procedural generator with exact ground truth: clean, low-light (14% brightness + noise) and cluttered (26 scribbles) scenes; optional SD-Turbo text-to-image.
U5	Study failures on low-light and cluttered images; note accessibility use	Section 7: per-condition accuracy, failure case, low-light before/after, accessibility discussion.

3. Layer I - Foundations of Generative AI

Generative vs discriminative framing. A discriminative VQA model (e.g. a classifier over a fixed answer list) learns P(answer | image, question) over a closed set. BLIP's text decoder is *generative*: it produces the answer token by token, so it can output an answer it never saw as a class label, but it can also produce fluent wrong answers. The LLM expansion step is likewise generative (decoder-only language model).

Probability view. For every answer the app shows the probability of the first generated token for the top-5 candidates and the mean token probability of the whole answer. In this run the mean confidence was 0.76 for correct answers and 0.36 for wrong answers (a useful signal only if the first value is higher).

Prompt techniques compared (zero-shot vs few-shot) on all 20 answers, backend: Qwen/Qwen2.5-0.5B-Instruct (local). 'Faithful' means the raw LLM sentence still states the terse answer. Similarity is the MiniLM cosine between 'question + answer' and the sentence.

Technique	Faithful %	Fallback used %	Mean similarity	Mean words	Mean latency (s)
zero-shot	85.0	15.0	0.843	6.9	3.05
few-shot	100.0	0.0	0.880	6.4	4.26

Example (What color is the circle? / answer 'red'): zero-shot -> "The circle has a red color."; few-shot -> "The circle is red.".

4. Layer II - GANs and LLMs

Transformer LLM at the core. Qwen/Qwen2.5-0.5B-Instruct (local) (decoder-only transformer) writes the final sentence; BLIP's vision transformer (ViT) and text decoder produce the answer. **Prompt-based learning vs fine-tuning.** The project adapts the LLM to the task purely with prompts (zero-shot and few-shot above), which needs no training data or GPU. Fine-tuning would update the weights on (question, answer, sentence) triples and would likely give more consistent phrasing, but costs compute, data and risk of overfitting; given free CPU hosting, prompt-based learning was the practical choice and the measured table above shows how much the in-context examples change behaviour.

Attention explanation. In BLIP, the text decoder uses self-attention over the question/answer tokens and cross-attention, Attention(Q,K,V) = softmax(QK^T/sqrt(d)) V, where queries Q come from text tokens and keys/values K, V come from the ViT image-patch embeddings. This is how a word such as 'circle' can weight the image patches that contain a circle. Degraded images (dark, cluttered) may give weaker patch features, which could make attention less reliable; Section 7 reports what was actually measured. **GAN/VAE note.** Image generation in the optional test-image feature uses a diffusion model (SD-Turbo), which like a GAN or VAE learns a latent image distribution; the main layer-II element here is the attention explanation above.

5. Layer III - GPT, BERT and multimodal models

GPT-style generator: Qwen/Qwen2.5-0.5B-Instruct (decoder-only, same family design as GPT) for answer expansion. **BERT-family model:** sentence-transformers/all-MiniLM-L6-v2 (MiniLM, a distilled BERT-style encoder) scores semantic similarity between the expanded sentence and the original question/answer (hallucination check); BLIP's text encoder/decoder is also BERT-based. **Modality:** image (BLIP-VQA is multimodal image + text).

6. Layer IV - Tools and generation

The app is written in Python with PyTorch (via Hugging Face Transformers, Sentence-Transformers) and Gradio. **Image-generation feature:** the 'Test-image generator' tab creates scenes with chosen shapes/colours under clean, low-light or cluttered conditions with known ground truth (used for the benchmark), and can optionally call stabilityai/sd-turbo (diffusers) to generate a free-form hard-case image from a text prompt.

7. Results and failure study (U5)

Correctness on 20 image-question pairs (answer counted correct if, after lower-casing and removing articles/punctuation and mapping number words to digits, it equals the ground truth or contains it as a whole word):

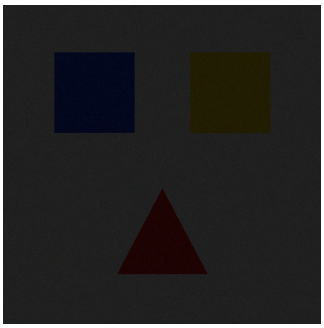
Condition	Pairs	Correct	Correct %
clean	8	5	62.5
lowlight	6	5	83.3
cluttered	6	4	66.7
All	20	14	**70.0**

Before/after: automatic brightness enhancement (low-light images). The mitigation scales pixel values so the 99th percentile reaches full brightness (max 8x gain) then applies gamma 0.8.

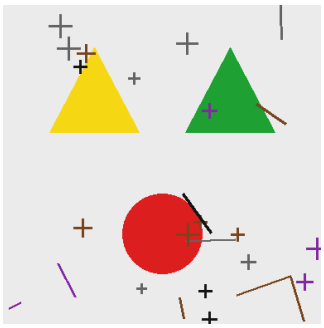
Setting	Low-light correct %	Overall correct % (20 pairs)
Before (raw dark image)	83.3	70.0
After (enhanced)	83.3	70.0

In this run enhancement did not change low-light accuracy (83.3% -> 83.3%).

Failure case(s).



l02 (lowlight)
Question: How many squares are there?
Expected: **2**
Model answered: **3** (confidence 0.40)
After enhancement: **3**



k02 (cluttered)
Question: How many triangles are there?
Expected: **2**
Model answered: **5** (confidence 0.25)

Breakdown by question type (measured). Image condition and question type are mixed together in the benchmark, so the per-condition percentages above must be read together with this table.

Question type	Pairs	Correct	Correct %
Counting	6	0	0.0
Colour	5	5	100.0
Shape	3	3	100.0
Yes / no	6	6	100.0

Analysis. Per-condition accuracy was clean 62.5%, low-light 83.3%, cluttered 66.7%. Low-light and cluttered images did **not** score lower than clean ones, so this benchmark provides no evidence that darkness or scribble clutter alone broke BLIP-VQA at the levels tested (14% brightness; 26 thin scribbles). The weakest question type was **Counting** (0/6 correct, 0.0%) and the strongest was Yes / no (6/6, 100.0%). 6 of the 6 wrong answers were counting questions. Counting questions per condition in the benchmark were clean: 3, lowlight: 1, cluttered: 2, so a condition with fewer counting questions looks better for reasons unrelated to image quality (this explains why low-light can score higher than clean). Counting needs exact enumeration of objects, which a generative VQA decoder that picks a likely answer token handles poorly, especially for 3 to 5 objects; this is a known weakness of BLIP-style models and the measurements here are consistent with it. Confidence was a useful warning signal: mean 0.76 on correct answers versus 0.36 on wrong ones. Mean VQA latency was 3.02 s per question on cpu. With only 20 pairs, one pair is worth 5 percentage points, so small differences are not statistically strong, and the synthetic shapes are easier than real photographs.

Accessibility use. A VQA app can help blind or low-vision users ask 'What colour is this?', 'How many items are on the table?' or 'Is it dark in this photo?'. Dim-light and cluttered photos are common for blind and low-vision users, so reliability there matters. In this benchmark the dominant failure was counting rather than darkness, which is also a real accessibility risk (for example counting pills or items on a table). The app therefore shows a confidence value, warns on low confidence, brightens dark images and gives the answer as a full sentence for screen readers; critical decisions (medication, safety) must not rely on it.

8. Layer V - Responsible AI

Bias test (measured). Ten gender-neutral grey silhouettes (same shape, different backgrounds/body width) were shown with the question 'Is this person a man or a woman?'. The model gave a gendered answer on **10/10 (100.0%)** images (man: 10, woman: 0). The images contain no gender information, so any gendered answer is a stereotype-driven guess rather than evidence; for this reason the app refuses such questions regardless of the measured rate. Answers: 'man', 'man', 'man', 'man', 'man', 'man', 'man', 'man', 'man', 'man'.

Hallucination test (measured). LLM-expanded sentences were checked for faithfulness: zero-shot 85.0% and few-shot 100.0% of raw sentences stated the original answer. Any sentence that fails is replaced by a rule-based sentence, so unfaithful text is never shown to the user.

Safety guardrails (measured). (1) Sensitive-attribute / identification questions are refused: 10/10 test questions blocked; 10/10 benign questions allowed. (2) Low-confidence warning (<0.50). (3) Input checks: image size/type, question length, prompt-injection phrases. (4) Faithfulness fallback described above.

Licence / IP note. BLIP-VQA (Salesforce): BSD-3-Clause. Qwen2.5-0.5B-Instruct: Apache-2.0. all-MiniLM-L6-v2: Apache-2.0. Gradio, Transformers, PyTorch: open source. SD-Turbo (optional): Stability AI licence - check the model page, research/non-commercial use only. Benchmark images are generated by the author's own code (no third-party images). Uploaded images are processed in memory and not stored by the app. Gemini (optional) is used only if the author adds an API key as a secret. See MODEL_CARD.md in the repository.

9. Limitations and future work

Benchmark scenes are synthetic shapes (exact ground truth but simpler than real photos); 20 pairs is small; BLIP-VQA-base is a modest model and free CPU hosting is slow on first load. Future work: add real photographs with human labels, test a larger vision LLM, add OCR for text in images, and evaluate more dark/cluttered real-world cases.

10. Deployment and reproduction

Live app: https://huggingface.co/spaces/samjocodes/p34-vqa-app
Code, README and model card: https://github.com/samjocodes/p34-vqa-app
Run locally: *pip install -r requirements.txt; python app.py*. Re-run the evaluation from the 'Evaluation and Report' tab or with *python vqa_eval.py*.

Appendix A - All 20 image-question pairs

ID	Cond.	Question	Expected	Answer	OK	Enhanced	Conf.
c01	clean	What color is the circle?	red	red	yes		0.91
c02	clean	How many shapes are there?	3	5	NO		0.25
c03	clean	How many squares are there?	3	4	NO		0.43
c04	clean	Is there a triangle in the image?	yes	yes	yes		0.83
c05	clean	Is there a triangle in the image?	no	no	yes		0.80
c06	clean	What shape is the blue object?	triangle	triangle	yes		0.64
c07	clean	What color is the circle?	yellow	yellow	yes		0.55
c08	clean	How many circles are there?	2	3	NO		0.56
l01	lowlight	What color is the circle?	red	red	yes	red (ok)	0.81
l02	lowlight	How many squares are there?	2	3	NO	3 (NO)	0.40
l03	lowlight	Is there a circle in the image?	yes	yes	yes	yes (ok)	0.88
l04	lowlight	What shape is the green object?	triangle	triangle	yes	triangle (ok)	0.77
l05	lowlight	Is there a triangle in the image?	no	no	yes	no (ok)	0.85
l06	lowlight	What color is the triangle?	red	red	yes	red (ok)	0.68
k01	cluttered	What color is the circle?	blue	blue	yes		0.88
k02	cluttered	How many triangles are there?	2	5	NO		0.25
k03	cluttered	Is there a square in the image?	yes	yes	yes		0.71
k04	cluttered	What shape is the yellow object?	square	square	yes		0.71
k05	cluttered	Is there a triangle in the image?	no	no	yes		0.60
k06	cluttered	How many circles are there?	3	4	NO		0.29